

Prototype – C · Complete Searchable Topic List

70 built-in visualisations, arranged subject by subject. Type the search term into the top bar and press Visualize. All of these work offline with no API key. Search tolerates spelling mistakes, and any equation such as $z = \sin(x)\cos(y)$ can be typed directly.

#	TOPIC	TYPE THIS IN THE SEARCH BAR	WHAT YOU SEE
LINEAR ALGEBRA & CALCULUS			
1	The derivative as a slope	derivative	Curve with a gold tangent you slide along it; live slope readout.
2	Integration as accumulated area	integration	Riemann rectangles fold into the curve as you raise the count; live sum.
3	Rolle's theorem	rolle theorem	Horizontal chord between equal endpoints; tangent must go flat at c.
4	Lagrange's mean value theorem	mean value theorem	Sloping chord plus tangent; the difference readout passes through zero at c.
5	Taylor series	taylor series	Polynomial terms added one at a time, creeping up on the true curve.
6	Changing the order of integration	change of order of integration	One region re-sliced vertically or horizontally; the limits rewrite themselves.
7	Volume under a surface	double integral	3D sheet over a domain; approximate volume updates with the spread.
8	Saddle point and partial derivatives	saddle point	Surface that is a minimum along x and a maximum along y.
9	A matrix as a transformation	matrix determinant	Grid deformed by a 2x2 matrix; determinant shown as the area scale factor.
10	Euler's formula and complex numbers	euler formula	Helix whose shadows are a sine and a cosine.
11	Fourier series	fourier series	Harmonics added until a square wave appears; Gibbs overshoot visible.
12	Polar curves	polar curves	Rose curves generated from r as a function of theta.
13	The normal distribution	normal distribution	Bell surface narrowing and sharpening as sigma changes.
DIFFERENTIAL EQUATIONS & VECTOR CALCULUS			
14	Direction field of an ODE	direction field	Arrow field showing the slope a solution must follow everywhere.
15	Divergence and curl	divergence curl	3D arrow field; separate sliders for swirl and outflow.
16	Green, Stokes and Gauss	stokes theorem	Interior circulation cancelling in pairs, leaving only the boundary.

#	TOPIC	TYPE THIS IN THE SEARCH BAR	WHAT YOU SEE
ENGINEERING PHYSICS - WAVES & MECHANICS			
17	Travelling wave	travelling wave	Moving sine with live wavelength, period, speed and frequency.
18	Electromagnetic wave	electromagnetic wave	E and B fields locked at right angles, propagating together.
19	Standing wave and harmonics	standing wave	Fixed-end string with nodes marked; harmonic number slider.
20	Two-source interference	ripple interference	Water-tank style 3D surface showing constructive and destructive lines.
21	Damped harmonic motion	simple harmonic motion	Oscillation inside a decaying envelope; damping slider.
22	Projectile motion	projectile motion	Parabolic path with live range, time of flight and max height.
23	Stress-strain curve	stress strain	Elastic region, yield point and plastic deformation.
24	Magnetic field around a current	magnetic field	Closed loops of field circling the conductor.
25	Rotational dynamics	moment of inertia	Bodies at different radii; outer ones lag badly behind.
26	Orbital motion	orbital motion	Kepler scaling - period grows with radius to the 3/2 power.
27	Exponential growth and decay	exponential decay	Half-life stays fixed no matter where you start.
28	The helix	helix	Circular motion plus steady drift; rise per turn readout.
ENGINEERING PHYSICS - OPTICS (lab view, not 3D)			
29	Newton's rings	newton rings	True fringe pattern with field stop, crosshair, scale bar and lens schematic.
30	Young's double slit	young double slit	Screen fringes with the single-slit envelope; live fringe width.
31	Single-slit diffraction	single slit diffraction	Central band twice the width of the side bands.
32	Diffraction grating	diffraction grating	White light split into orders - real spectral colour.
33	Air-wedge fringes	air wedge	Straight, evenly spaced fringes of equal thickness.
34	Step-index and graded-index fibre	graded index fibre	Cross-section with ray path; live NA and acceptance angle.
35	Light trapped in a fibre	total internal reflection	Ray zig-zagging down the core past the critical angle.
SEMICONDUCTOR PHYSICS (ECE)			

#	TOPIC	TYPE THIS IN THE SEARCH BAR	WHAT YOU SEE
36	P-N junction and depletion region	pn junction	Depletion layer widens and narrows with bias; carriers flow when forward.
37	Diode V-I characteristic	diode characteristics	The knee in forward bias, flat leakage in reverse.
38	Hall effect	hall effect	Carriers deflected to one face; live Hall voltage and coefficient.
BASIC ELECTRICAL & ELECTRONICS (BEEE)			
39	AC phasors and power factor	ac circuits	Rotating phasors beside the matching waveforms; live impedance and cos phi.
40	Single-phase transformer	transformer	Core with flux; turns ratio sets the secondary voltage live.
41	Half-wave rectifier	half wave rectifier	Circuit plus input and output waveforms side by side.
42	Full-wave bridge rectifier	bridge rectifier	Four-diode bridge; negative half flipped up instead of lost.
43	BJT common-emitter amplifier	common emitter	Full bias circuit; collector voltage falls as base current rises.
44	Common-base configuration	common base	Grounded base, alpha just under one, no phase inversion.
45	Inverting op-amp	operational amplifier	Virtual earth marked; gain is nothing but the resistor ratio.
NETWORK ANALYSIS (ECE core)			
46	Thevenin and Norton equivalents	thevenin theorem	Both forms side by side feeding the same load with the same current.
47	Series resonance and Q factor	series resonance	Resonance peak with half-power points; live bandwidth and cutoffs.
48	RL and RC transients	time constant	Charging and discharging exponentials; 63% at one tau.
49	Two-port network parameters	two port network	V1, I1, V2, I2 at the ports; Z, Y, ABCD and h sets explained.
ENGINEERING CHEMISTRY (ECE stream)			
50	The water molecule	water molecule	Bent geometry at 104.5 degrees - why water is polar.
51	sp ³ hybridisation	sp3 hybridization	Methane's tetrahedron, four identical bonds at 109.5 degrees.
52	Benzene	benzene	Delocalised ring; every C-C bond identical.
53	Ionic crystal lattice (NaCl)	crystal lattice nacl	3x3x3 rock-salt block - no such thing as one NaCl molecule.
54	A p orbital	p orbital	Two lobes with a node between them, drawn in spherical coordinates.
55	Crystal field splitting	crystal field splitting	Octahedral and tetrahedral d-orbital splitting; live delta and CFSE.

#	TOPIC	TYPE THIS IN THE SEARCH BAR	WHAT YOU SEE
56	Ion-exchange water softening	ion exchange	Resin column with beads; saturation level and hardness removed.
57	Galvanic corrosion cell	galvanic corrosion	Anode dissolving, electrons through the wire, a pit forming; live EMF.
PROGRAMMING & COMPUTING			
58	Loops and memory	for loop flowchart	Flowchart beside live memory cells; the counter slot overwritten each pass.
59	Program control flow	program control flow	Execution path through sequence, condition and loop.
60	Recursion	recursion	Calls descending to the base case, then answers bubbling back up.
61	Bubble sort	bubble sort	Animated bars with the current comparison highlighted; step counter.
62	Binary search tree	binary search tree	In-order walk highlighting nodes in sorted sequence.
63	Linked list	linked list	Sequential access - reaching item k costs k steps.
64	Big-O growth rates	big o complexity	$\log n$, n , $n \log n$ and n^2 compared on one axis.
65	A neural network	neural network	Layered graph with a forward pass sweeping left to right.
66	The design thinking cycle	design thinking	Five stages as a loop, because testing sends you back.
BASIC CIVIL & MECHANICAL ENGINEERING			
67	Shallow and deep foundations	shallow foundation	Soil cross-section; spread footing vs pile with shaft friction and end bearing.
68	Four-stroke engine cycle	four stroke engine	Piston and valves animating through all four strokes, labelled.
69	Two-stroke engine cycle	two stroke engine	Ports uncovered by the piston skirt; power every revolution.
70	Vapour compression refrigeration	vcrs	Refrigerant circulating through all four components; live COP.

Also: any equation can be typed straight into the search bar — for example $z = \sin(x)\cos(y)$ or $a\exp(-b*x)\cos(4*x)$ — and it is plotted directly. Search tolerates spelling mistakes. **Controls:** drag to orbit, scroll or pinch to zoom, Space pauses, R resets the view.